Additional file 1: Supplementary Material

Evaluation area and the limits of wildfire model transfer

Supplementary material for *Evaluation area and the limits of wildfire model transfer*.

The figures carry most of the results, and the text adds only what they do not show. Every value is read from a released output of the repository https://github.com/dryuemco/thermal-twin. File paths that do not start with `paper/` are under `paper/labelfix_rerun/`, which holds the corrected-label outputs, and values computed under the original Manavgat label are marked "original label". The script that draws each figure checks every plotted value against its source, and `paper/code/appendix_tables.py` rebuilds Tables S2 to S4 and S7 to S11 row by row.

Contents

- S1 Robustness checks
  - S1.1 The within-region gain
  - S1.2 Cross-region transfer
  - S1.3 Resampling units
  - S1.4 Further checks
- S2 The evaluation frame within a region
  - S2.1 Scar by scar
  - S2.2 Distance from the training cells
- S3 Signed associations and the frame test
  - S3.1 Signed associations and reversals
  - S3.2 The 10 km collar
  - S3.3 The temperature sign with terrain held
  - S3.4 The contrast pair
  - S3.5 The two Muğla fires
- S4 Transfer, adaptation and similarity
  - S4.1 The transfer matrix
  - S4.2 The transfer-gap decomposition
  - S4.3 Precision and recall
  - S4.4 Pooled training
  - S4.5 Similarity measures
- S5 Data and methods
  - S5.1 Label, grid, gate and the Manavgat correction
  - S5.2 Study areas and windows
  - S5.3 Predictors
  - S5.4 Resampling units
  - S5.5 Gap decomposition and reversal criterion
  - S5.6 Label-free adaptation
  - S5.7 Reproduction, code and data

S1 Robustness checks

S1.1 The within-region gain

Fig. S1 sets the within-region thermal gain under each check beside the gain it was compared with. The gain stays above zero in every region under every check. The only interval that includes zero is that of Montiferru with aspect added, +0.020 [−0.025, +0.066], against +0.099 without aspect. On the 10 km collar the mean over the five regions is +0.083 [+0.037, +0.129], against +0.087 [+0.037, +0.136] as drawn (Student *t* over regions); on the 5 km collar it halves to +0.042.

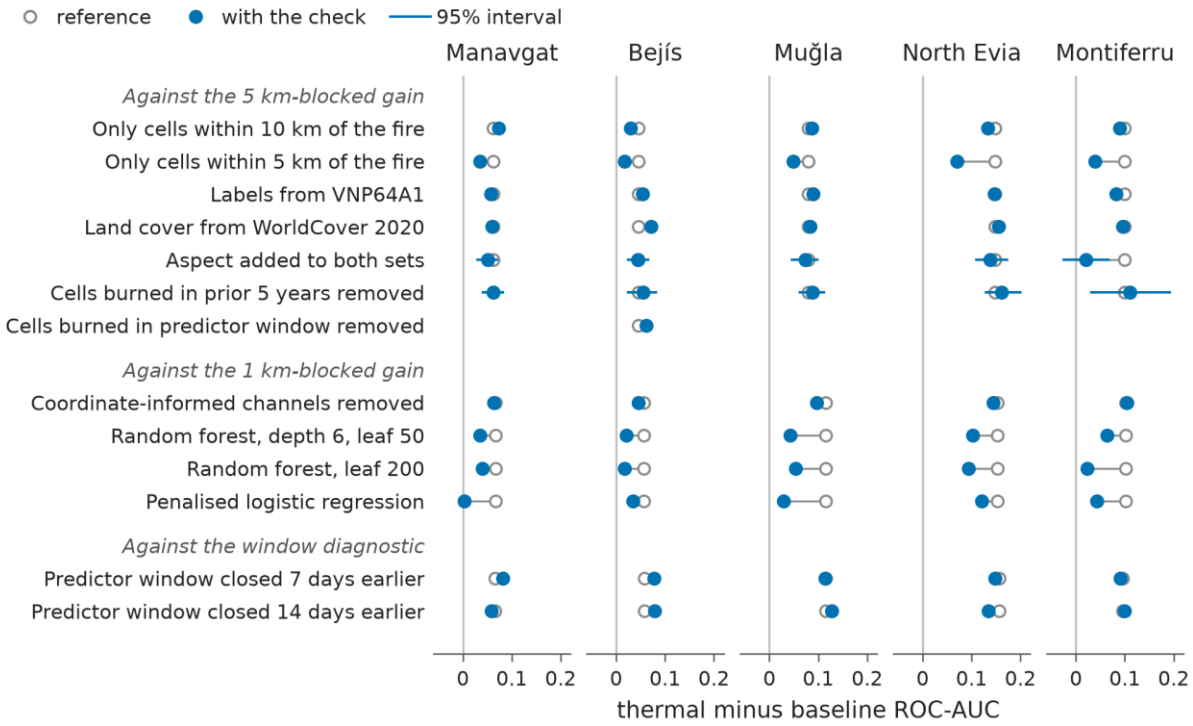


Fig. S1. The within-region thermal gain under each check, by region. *Thermal minus baseline ROC-AUC. Open dots are the reference that a check is compared with (italic headings): the 5 km-blocked gain, the 1 km-blocked gain of the random forest of the main text, or the pipeline's own window diagnostic. Filled dots are the gain with the check, and lines are 95 % block-bootstrap intervals where the source gives one. Drawn by* `paper/figures/figS1_within_checks.py` *from* `inference/ladder_summary.json`*,* `round8/` *(r8b, r8j, r8m, r8n),* `labels/r2_history.json`*,* `round3/no_coord_channels.json`*,* `code/model_capacity.json` *and* `exports/window_closure_<region>/`*.*

A few details are not in the figure. VNP64A1 and MCD64A1 labels agree at a Jaccard index of 0.865 (Montiferru) to 0.978 (Evia). With WorldCover 2020, at most 43 burned cells leave or enter the natural-vegetation population in four regions, but 142 enter it in Montiferru, where the share of burned cells with at least half natural vegetation rises from 0.773 to 0.977 (on the modelled cells; the gate fraction of Table 1 comes from the gate's own inputs). In Bejís, 48 cells burned inside the predictor window and ten of them are labelled burned; removing them changes transfer in the eight Bejís directions by at most 0.015. Rebuilding the Manavgat chain from a quality-screened MODIS input moved the gain from +0.067 to +0.068 and no signed association by more than 0.005 (`round6/qc/qc_compare_manavgat_corrected.json`), although the two versions also differ in pipeline code, because the current step7 refuses the unscreened raster. The secondary population of all valid cells (Section 2.5), which exists only in Manavgat and Bejís, gives gains of +0.048 to +0.061 over the two regions and three block sizes, every interval above zero, against +0.045 to +0.067 in the primary population (`pipeline/robustness/step8_large_block_primary_all_valid/`).

S1.2 Cross-region transfer

Fig. S2 repeats the transfer analysis under each check. Mean transfer ranges from 0.481 (penalised logistic regression) to 0.566 (the twelve directions without Manavgat), against 0.527 as drawn, and every interval of

the thermal gain includes zero. The gain is +0.007 for the random forest of the main text and negative for the three other classifiers. Without Manavgat, mean transfer on the 10 km collar is 0.631, with no direction below chance (`round8/r8d_summary.json`).

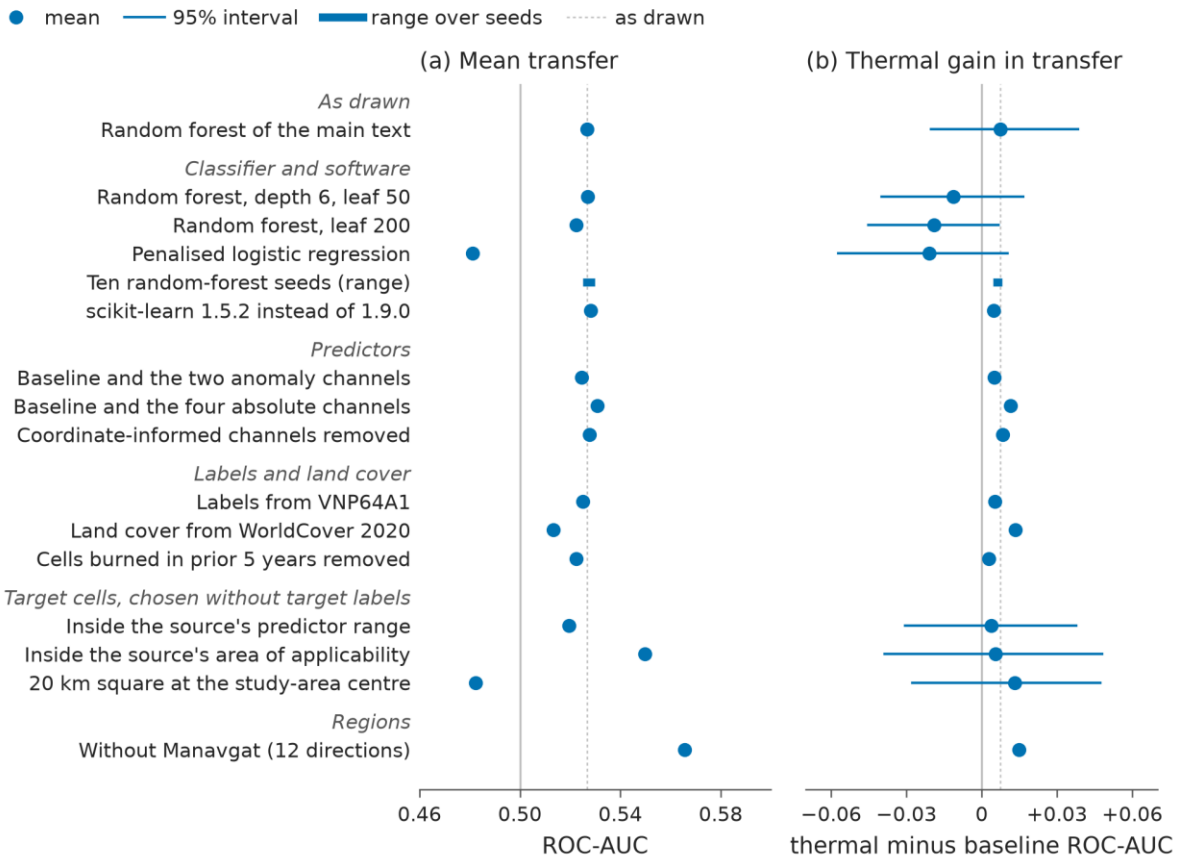


Fig. S2. Mean cross-region transfer under each check. *(a) Mean thermal ROC-AUC over the twenty directions (twelve without Manavgat). (b) Mean thermal minus baseline ROC-AUC; for the three predictor sets it is taken against the baseline-only model (0.519). Lines are pair-cluster 95 % intervals where computed, and thick bars are ranges over ten random-forest seeds. The last three target frames use no target label: cells inside the source's predictor range or area of applicability (Meyer and Pebesma 2021), or a 20 km square at the centre of each study area. Drawn by* `paper/figures/figS2_transfer_checks.py`*, which lists its sources.*

The anomaly channels and the absolute channels transfer about equally. Per direction their difference averages −0.006, is positive in 11 of 20 directions and ranges from −0.099 to +0.087, and within regions they reach 0.862 and 0.864, between the baseline (0.797) and all ten predictors (0.896). Between the two Muğla fires the anomaly channels keep their direction and the absolute channels do not (Section S3.5), but `lst_anomaly_mean` reverses between regions (Table S3), so normalisation protects a channel against a change of season in one place and fails across places.

Removing elevation and the LST anomaly together (Fig. 7) lowers mean within-region AUC by 0.076, with every regional interval below zero (−0.035, −0.130, −0.073, −0.063 and −0.079 for Manavgat, Bejís, Muğla, Evia and Montiferru), and changes mean transfer by +0.014, with a pair-cluster interval of [−0.018, +0.049] (`round8/r8q_summary.json`). About three quarters of the within-region cost comes from elevation: without it alone the mean falls from 0.896 to 0.840, and without the LST anomaly alone to 0.883.

On the earlier, smaller Evia rectangle, whose burned prevalence is higher, the raw transfers between Evia and Bejís, Manavgat and Muğla moved by up to 0.134 (Evia to Manavgat, 0.543 against 0.677), and none changed side of chance (`round7/r7f_evia_legacy.csv`).

CORAL hardly depends on its regularisation. Over nine values of λ from 0 to 10^{-1} on four directions, transfer moved by at most 0.014 (`exports/coral_lambda_sensitivity_b74d643e/`), and the Manavgat and Bejís pair, whose adapted result depended on λ under the original label, stays below chance in both

directions at every λ from 10^{-5} to 1 (`step10/coral_lambda_sensitivity.csv`). CORAL with λ = 1 gives a mean of 0.512 and CORAL on two thermal principal components 0.523. A placebo that aligns each source with a third region instead of the target gives 0.507 and moves 15 of 20 directions toward chance, against 16 of 20 for CORAL, so this movement is not specific to the target (`round8/r8c_summary.json`).

S1.3 Resampling units

Fig. S3 gives the interval of the paired thermal contribution under every resampling unit that can be computed. As drawn, every interval includes zero. On the 10 km collar only the clusters by target region, [+0.010, +0.040], and by source region, [+0.010, +0.042], exclude it, and with five regions a percentile bootstrap over regions has at most 126 distinct resamples, so both are coarse. The two-way cluster and dyadic-robust estimators give negative variances and are undefined. In an equivalence test the gain as drawn lies within ±0.05 under all six computable units (90 % upper bounds 0.021 to 0.037) and within ±0.02 under none; on the collar it lies within ±0.05 under four of the six, the exceptions being the pigeonhole bootstrap and the region jackknife (upper bounds 0.052 and 0.055). Leaving out one region at a time moves the mean contribution between +0.0010 (without Evia) and +0.0148 (without Manavgat), and its sign never changes (`code/transfer_delta_ci.json`).

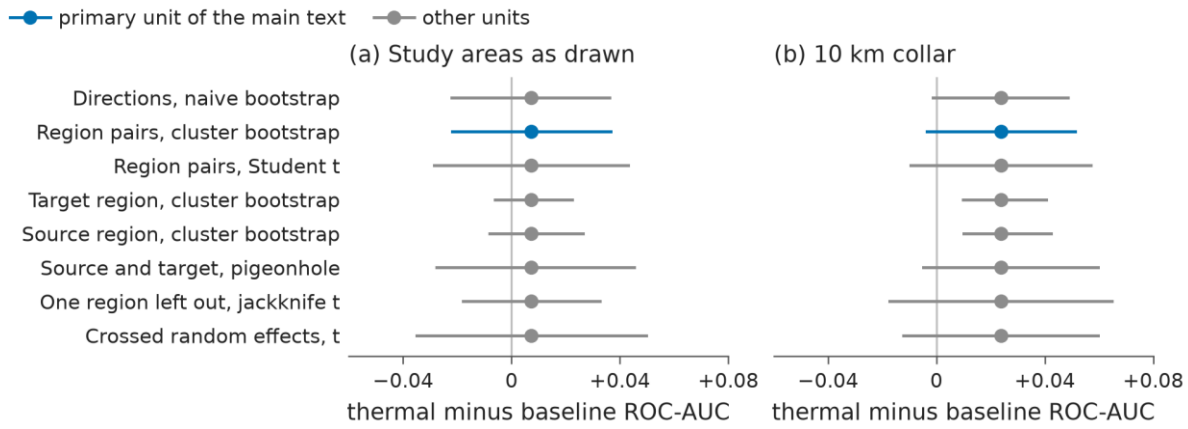


Fig. S3. The paired thermal contribution to transfer, by resampling unit. *Mean thermal minus baseline ROC-AUC over the twenty directions (dot) and its 95 % interval (line), (a) on the study areas as drawn and (b) on the 10 km collar; 20,000 replicates where resampled. Blue is the pair-cluster bootstrap, the primary unit of the main text, whose interval as drawn is [−0.020, +0.038] at the 1000 replicates used there. Drawn by* `paper/figures/figS3_units.py` *from* `round8/r8d_summary.json`, `inference/units.json`, `code/transfer_delta_ci.json` *and* `round8/r8l_summary.json`.

S1.4 Further checks

Table S1 lists the checks that are not plotted. Like those of Figs. S1 to S3, they were run after the main results, and none was used to choose a reported configuration.

Table S1. Further checks. Sources in `geometry/` (a1 to a5) and `round8/` (R8 labels).

Check	Result
Scar edge (a1)	Excluding 0, 1 or 2 edge cells from the scar frame leaves the frame cost (A − B, nine scars) at 0.147 [0.086, 0.208], 0.137 [0.069, 0.205] and 0.129 [0.055, 0.204]; with the region as unit 0.160, 0.150 and 0.145, every interval above zero
Placebo collars (a2)	The burned cells of a scar scored against the unburned cells of a same-shaped collar placed away from any burned cell reproduce A (A − placebo −0.002 [−0.056, +0.052], eight scars); the real collar costs +0.155 [+0.108, +0.202] more
Distant cells (a3)	The frame cost does not grow with the share of cells beyond 10 km: region-level slope 0.005 [−0.41, +0.42], r = 0.02 (n = 5); scar-level slope −0.049 [−0.34, +0.24] (n = 9, scars not independent)
Other metrics (a5)	Region-wide minus scar frame over nine scars: partial AUC (FPR ≤ 0.1) 0.088 [0.048, 0.128]; average precision −0.342 [−0.477, −0.206], higher on the scar frame because its prevalence is higher

Check	Result
Ten forest seeds (R8a)	The seed-to-seed standard deviation of a direction has a median of 0.005 (maximum 0.015), and no direction changes side of chance as drawn; on the 10 km collar one does (Montiferru to Manavgat). Averaging the ten forests gives 0.527 (gain +0.006) as drawn and 0.590 (+0.025) on the collar
Capture of burned cells (R8b, R8g)	In transfer, the 10 % of target cells with the highest thermal scores hold 10.3 % of the burned cells on average (0.7 % to 27.1 %), as a random ranking would, and eight of twenty directions fall below that; the top 20 % hold 19.9 %. Within regions (5 km blocking) the top 10 % hold 35.5 % (18.9 % to 44.5 %) and the top 20 % 55.7 %
Weather reference (R8o)	Across the five regions the mean Landsat LST anomaly follows the ERA5-Land air-temperature anomaly of the predictor window (Spearman $\rho = 0.8$) but not the relative-humidity anomaly ($\rho = 0.0$), and neither does the TVDI difference ($\rho = 0.0$); with five regions this is descriptive
Missing thermal values (R8p)	1.7 %, 6.5 %, 3.2 %, 4.5 % and 0.1 % of the modelled cells (Manavgat, Bejís, Muğla, Evia, Montiferru) lack at least one thermal channel; this flag scores 0.493 to 0.512 against burning
Residual spatial correlation (R8l)	Out-of-fold residuals of cell pairs correlate at 0.58 to 0.83 below 1 km and 0.15 to 0.48 at 5 to 10 km (Montiferru −0.01); the correlation falls below 0.05 at 10 to 20 km in Manavgat, Bejís and Muğla, at 20 to 40 km in Evia and at 5 to 10 km in Montiferru, so 5 km intervals may be too narrow, most of all in Evia
Pooled against fold AUC (R8l)	The mean of the five fold AUCs is 0.004 to 0.018 above the pooled out-of-fold AUC in every region (thermal model, 5 km blocking)

S2 The evaluation frame within a region

S2.1 Scar by scar

Fig. S4 shows the evaluations of Section 3.3 for each scar. Rows A to D of Table 3 average the seven scars that have a row C, four in Muğla, two in Montiferru and one in Evia; Manavgat and Bejís each have one burned area, and withholding it would leave nothing to train on. The controls reuse the predictions of A and are averaged over 20 draws without replacement. The prevalence-matched sample draws $|H_K^+|$ burned and $|H_K^-|$ unburned region cells, the negative-pool swap keeps the drawn burned cells and takes the unburned cells of the scar, H_K^-, and the positive-pool swap does the opposite.

Over nine scars the prevalence-matched sample (A′) scores 0.793 against 0.791 for the whole region, as ROC-AUC should, so the fall to 0.644 on the scar area (B) is not an effect of prevalence. Replacing only the unburned cells of the sample with those of the scar collar costs 0.139 [0.098, 0.181], and replacing only the burned cells −0.002 [−0.055, +0.051] (`code/pool_decomposition.json`): the cost comes from unburned cells next to the fire, which share its terrain, land cover and weather. The intervals of Table 3 are Student *t* intervals over scars, and row A repeats one region value for each scar of a region, so its intervals and those of A − B and A − C are too narrow. With the region as the unit, A − B is +0.137 [+0.048, +0.226] over three regions and +0.160 [+0.090, +0.230] over all nine scars in five regions, but A − C is +0.266 [−0.022, +0.553]; only the frame cost is established (`round7/round7_summary.json`). Row C depends little on how a scar is defined: over a range of minimum sizes and buffers it ranges from 0.541 to 0.561 on nine to six scars (`code/scar_definition_sweep.csv`). For a single scar, row D depends on which region is the source; over the 36 source-scar combinations the mean is 0.556, and ten are below chance.

Over the seven scars, the thermal gain at 5 km blocking is +0.095 on the whole region, +0.021 on the scar area, +0.024 with the scar withheld and +0.008 from a foreign region. Scar area minus withheld is −0.004 [−0.070, +0.063] (+0.031 [−0.027, +0.090] at 1 km), and whole region minus scar area +0.074 [−0.009, +0.157]. With three regions no reliable region-level interval exists, and effects of up to about 0.2 are not excluded (`inference/ladder_summary.json`). A half-split of each region gives a gain of +0.028, positive in 13 of 18 splits, but a straight cut leaves unequal numbers of burned cells in the two halves (`code/positive_control.json`).

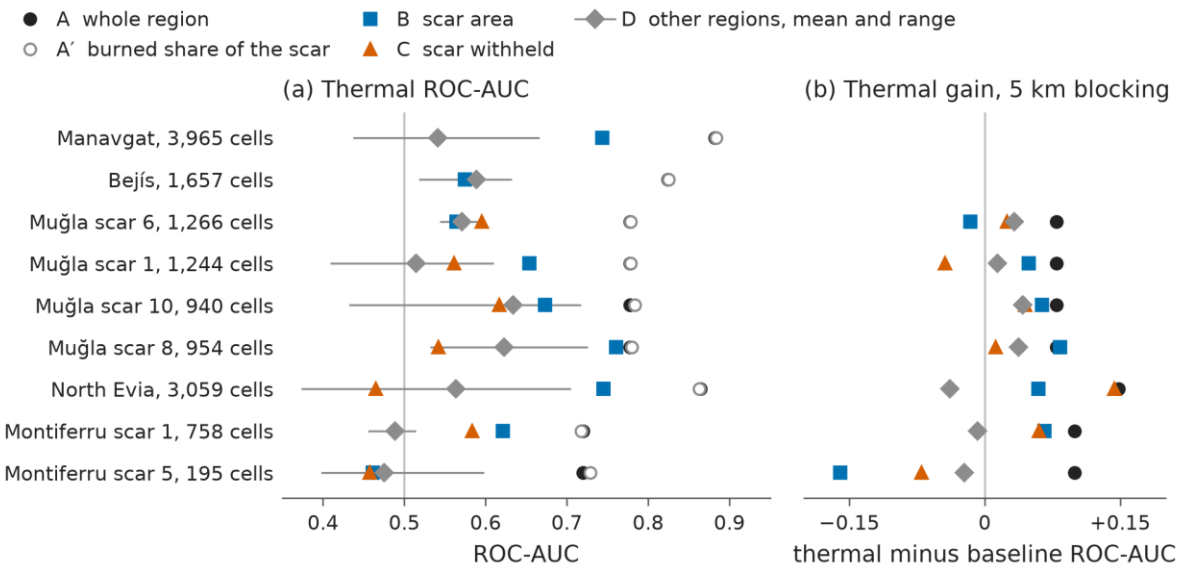


Fig. S4. The evaluations of Section 3.3, scar by scar. *(a) Thermal ROC-AUC of one region model scored on the whole region (A), on region cells with the burned share of the scar area (A') and on the scar area (B); of a model fitted without the scar (C); and of the models of the other four regions (D, mean and range). (b) Thermal minus baseline ROC-AUC under A, B, C and D for the seven scars with a row C. Natural-vegetation population, 5 km blocking, 2 km collar. Drawn by* `paper/figures/figS4_scars.py` *from* `code/matched_holdout.json`, `code/prevalence_control.json`, `code/d_per_source.json` *and* `inference/ladder_summary.json`.

S2.2 Distance from the training cells

A model fitted on one half of a region scores the other half, with the target cells binned by their distance from the training cells (`paper/code/distance_curve.py`). Skill falls to near chance at 10 to 20 km (Fig. S5), which limits how far from its training cells such a model can be used. The far bins hold few cells, so the rise at 40 to 80 km is not evidence of an effect, and the near bins keep the spatial autocorrelation that blocked validation removes, so 0.709 is an upper bound. The two distance ranges do not overlap (2 to 86 km within regions, 306 km and more across them), and distance is not separated from the land cover, terrain and fire history that change with it, so the curve gives a length scale and no cause. Once the curve reaches chance, any cross-region mean near 0.5 lies on its continuation, so this comparison cannot show that the cross-region failure is only an effect of distance. A model without information gives no reliably reversed ranking, yet Manavgat to Bejís is below chance with interval support on both frames. A leave-one-block-out reference with a 10 km buffer was tried and is not reported, because its pooled AUC mixed models with different calibration.

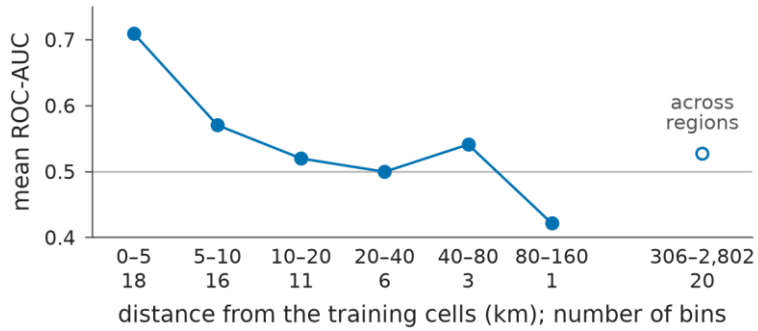


Fig. S5. Within-region skill by distance from the training cells. *Mean target ROC-AUC, unweighted over the distance bins of the five regions; the number of bins is under each class, and a bin holds 1 to 1,067 burned cells. The open dot is the mean of the twenty cross-region directions, 306 to 2,802 km apart. Drawn by* `paper/figures/figS5_distance.py` *from* `round7/r7h_distance_curve.csv`.

S3 Signed associations and the frame test

S3.1 Signed associations and reversals

Table S2 gives the signed association of each predictor with burning in each region, and Table S3 the pairs of regions that reverse under the strict criterion of Section 2.10: the two associations lie on opposite sides of 0.5 and the interval of each region excludes 0.5. This is stricter than disjoint intervals. For `current_lst_mean` the Manavgat and Bejís intervals are disjoint ([0.608, 0.719] against [0.401, 0.547]), but the Bejís interval includes 0.5, so the pair reverses at the point estimate only.

Table S2. Signed univariate AUC of each predictor against `burned`, by region. Natural-vegetation population; 10-cell (~5 km) spatial-block bootstrap, 1000 replicates, seed 42. The AUC is not folded to max(AUC, 1 − AUC), so a value below 0.5 means that lower values rank burned cells higher. Bold marks a region whose own interval excludes 0.5. Source `round5/tables/step9g_multi_aoi_feature_stability.csv` (the pipeline's Step9G synthesis, sha256 c864cd7d…).

Feature	Manavgat	Bejís	Muğla	Evia	Montiferru
`elevation_mean`	**0.232** [0.179, 0.288]	**0.643** [0.558, 0.729]	**0.611** [0.532, 0.690]	0.541 [0.448, 0.626]	0.584 [0.395, 0.762]
`slope_mean`	**0.400** [0.340, 0.459]	0.521 [0.439, 0.605]	**0.637** [0.582, 0.686]	0.487 [0.418, 0.554]	**0.652** [0.506, 0.771]
`ndvi_mean`	0.564 [0.499, 0.628]	0.559 [0.497, 0.619]	**0.662** [0.616, 0.704]	**0.639** [0.575, 0.701]	0.586 [0.450, 0.704]
`lst_anomaly_mean`	0.509 [0.460, 0.560]	**0.418** [0.364, 0.480]	0.485 [0.395, 0.566]	**0.640** [0.567, 0.710]	0.395 [0.285, 0.535]
`current_lst_mean`	**0.665** [0.608, 0.719]	0.477 [0.401, 0.547]	**0.325** [0.271, 0.382]	**0.377** [0.301, 0.456]	0.370 [0.248, 0.513]
`current_tvdi_mean`	**0.677** [0.622, 0.733]	0.517 [0.429, 0.595]	**0.336** [0.275, 0.398]	**0.362** [0.285, 0.442]	**0.356** [0.233, 0.499]
`tvdi_difference_mean`	0.460 [0.409, 0.510]	0.512 [0.443, 0.583]	0.490 [0.396, 0.575]	0.519 [0.444, 0.589]	**0.378** [0.282, 0.497]
`downscaled_lst_mean`	**0.683** [0.626, 0.739]	0.484 [0.400, 0.560]	**0.307** [0.253, 0.366]	**0.377** [0.297, 0.459]	0.365 [0.240, 0.511]
`fused_lst_mean`	**0.666** [0.610, 0.721]	0.481 [0.404, 0.551]	**0.325** [0.272, 0.383]	**0.376** [0.300, 0.456]	0.370 [0.248, 0.513]

Table S3. Cross-region reversals that meet the strict criterion. From the intervals of Table S2. Difference intervals from the paired 10-cell block bootstrap of `paper/code/ems_inference_multiplicity.py` (`inference/reversal_family_holm.csv`, 1000 replicates), which flags the same fourteen pairs.

Feature	Region A	AUC	Region B	AUC	Difference [95 % CI]
`elevation_mean`	Manavgat	0.232	Bejís	0.643	+0.411 [+0.312, +0.509]
`elevation_mean`	Manavgat	0.232	Muğla	0.611	+0.379 [+0.274, +0.476]
`slope_mean`	Manavgat	0.400	Muğla	0.637	+0.237 [+0.158, +0.313]
`slope_mean`	Manavgat	0.400	Montiferru	0.652	+0.252 [+0.100, +0.389]
`lst_anomaly_mean`	Bejís	0.418	Evia	0.640	+0.222 [+0.129, +0.316]
`current_lst_mean`	Muğla	0.325	Manavgat	0.665	+0.340 [+0.260, +0.413]
`current_lst_mean`	Evia	0.377	Manavgat	0.665	+0.288 [+0.195, +0.384]
`current_tvdi_mean`	Muğla	0.336	Manavgat	0.677	+0.341 [+0.253, +0.422]
`current_tvdi_mean`	Evia	0.362	Manavgat	0.677	+0.315 [+0.221, +0.413]
`current_tvdi_mean`	Montiferru	0.356	Manavgat	0.677	+0.322 [+0.161, +0.453]
`downscaled_lst_mean`	Muğla	0.307	Manavgat	0.683	+0.376 [+0.295, +0.451]
`downscaled_lst_mean`	Evia	0.377	Manavgat	0.683	+0.307 [+0.209, +0.405]
`fused_lst_mean`	Muğla	0.325	Manavgat	0.666	+0.341 [+0.261, +0.413]
`fused_lst_mean`	Evia	0.376	Manavgat	0.666	+0.291 [+0.198, +0.386]

Thirteen of the fourteen reversals involve Manavgat. Under the original label there were three, on elevation and the LST anomaly, the two predictors that the feature-removal analysis of Section 2.11 keeps. Twenty-six further pairs reverse at the point estimate only. None of the fourteen is corrected for multiplicity; Table S5 gives the corrected counts.

S3.2 The 10 km collar

The collar keeps every burned cell and the unburned cells within 10 km of one, so it removes distant unburned cells and nothing else (Fig. 1b). In Muğla one burned area touches the eastern edge of the study area (15 burned cells on the edge), where its collars are cut. The interval criterion was applied at 10 km, and the 5 and 10 km radii agree at the point estimate. The collar removes many cells in two regions, so part of any lost support is lost power; on average 2.3 features per direction are supported in both regions as drawn and 2.2 on the collar. Table S4 compares the three associations that straddle 0.5 on both frames.

Table S4. Signed univariate AUC as drawn and on the 10 km collar. Point estimates, signed as in Table S2; the collar removes no burned cell. As drawn from the source of Table S2, collar from `code/aoi_frame_auc_frozen_mugla.csv`, with 10-cell intervals in `code/collar_frame_bootstrap.csv`. On every row Manavgat lies on the other side of 0.5 from the other regions.

Signed univariate AUC	Manavgat	Bejís	Muğla	Evia	Montiferru	straddles 0.5
elevation, full frame (Table S2)	**0.232**	0.643	0.611	0.541	0.584	**yes**
elevation, 10 km collar	**0.376**	0.614	0.606	0.648	0.581	**yes**
current LST, full frame	**0.665**	0.477	0.325	0.377	0.370	**yes**
current LST, 10 km collar	**0.522**	0.405	0.332	0.286	0.376	**yes**
current TVDI, full frame	**0.677**	0.517	0.336	0.362	0.356	**yes**
current TVDI, 10 km collar	**0.527**	0.454	0.342	0.250	0.361	**yes**

Table S5 counts how many of the ninety feature-by-pair comparisons of each frame pass each test. The per-comparison criterion leaves two reversals on the collar, both on elevation and both involving Manavgat: against Muğla, also supported as drawn, and against Evia, supported only on the collar. Manavgat against Bejís misses support on the collar by 0.004. Neither collar reversal survives the Holm intersection-union correction (adjusted $p = 0.33$ against Evia and 0.79 against Muğla, normal approximation).

Table S5. Reversal tests, ninety feature-by-pair comparisons per frame. Source `inference/reversal_family_holm.csv` and `inference/reversal_per_region.csv`.

Criterion	Full frame	10 km collar
Difference interval excludes zero	39	27
… with opposite-sided point estimates	31	21
Per-comparison criterion (each region's own interval excludes 0.5, opposite sides)	14	2
Holm, normal approximation, difference test	27	12
Holm intersection-union (both regions' tests)	5	0

With only the target or only the source restricted to the collar, mean transfer is 0.559 and 0.546 (`round5/collar/aoi_frame_transfer.csv`); Table 4 gives the matched frames. On these frames the within-region reference at 5 km blocking is 0.814 as drawn, 0.786 on the 10 km collar and 0.759 on the 5 km collar (`paper/code/verify_matched_gap.py`), so transfer falls short of it by 0.287, 0.197 and 0.169; paired by target region the 10 km shortfall is +0.197 [+0.091, +0.303].

Of the twenty similarity measures, the two built from signed associations were recomputed on the collar. There the all-feature cosine orders transfer at $\rho = +0.57$ [+0.05, +0.88], the only interval of a similarity measure in the study that excludes zero (`round6/diag_collar/`, `round7/r7g_collar_diagnostics.json`). We do not use it: it is one of about forty uncorrected tests, it rests on twenty directions that share regions, and it needs target labels twice, because the collar is drawn from the burned cells.

After the frozen tables were computed, a quality-screening rebuild overwrote the Manavgat and Muğla modelling datasets at the pipeline path with new values of `downscaled_lst_mean` and `fused_lst_mean`. Every analysis of Section 3.4 therefore reads each region through `paper/code/_canonical.py`, which checks the SHA-256 of the frozen copy, and reads Manavgat from the corrected re-freeze (`5a5e876c…`).

S3.3 The temperature sign with terrain held

Section 3.4 reports a negative association between pre-fire surface temperature and burning in four of five regions, and the same in Manavgat once elevation is held (Table S6). In Manavgat the raw association is above 0.5 and falls below it within elevation deciles, 0.403 [0.343, 0.464] on the collar, and after removing the linear effect of elevation (−5.6 K per km), 0.409 [0.350, 0.475]; on the original study area the same values are 0.455 [0.396, 0.509] and 0.446 [0.387, 0.502], which include 0.5 (`round8/r8e_stratified_intervals.csv`). The positive raw signal of Manavgat therefore comes from terrain. In the other four regions the sign stays below 0.5 within NDVI deciles in all four and within elevation deciles in three. NDVI reverses in two regions once LST is held, while LST reverses in none once NDVI is held. Holding northness through deciles leaves the sign unchanged in every region: 0.652, 0.482, 0.317, 0.376 and 0.417 against 0.665, 0.478, 0.324, 0.377 and 0.370 raw (with missing values filled by the median, so the raw values differ from Table S2 in the third decimal; `round8/r8m_summary.json`). In Manavgat, burn timing and elevation are confounded: cells that burned in the first four days lie at a median of 219 m, later burns at 512 m and unburned cells at 1,004 m.

Table S6. The LST association with one variable held, 10 km collar. Signed AUC against `burned`. The held columns pool the concordance within deciles of the named variable, each decile weighted by its burned-unburned pairs; “detrended” removes the linear dependence of LST on elevation in the region. Bold marks the values discussed in the text. Sources `code/matched_frame_gap.csv` (recomputed by `paper/code/verify_matched_gap.py`) and `round7/r7b_lst_given_terrain.csv`; intervals for every cell in `round8/r8e_stratified_intervals.csv`.

Region	LST raw	LST within elevation	LST detrended on elevation	LST within NDVI	LST within distance	NDVI raw	NDVI within LST
Manavgat	0.522	**0.403**	**0.409**	0.556	0.454	0.551	0.550
Bejís	0.405	0.509	0.411	0.486	0.350	0.618	0.574
Muğla	0.332	0.363	0.365	0.404	0.367	0.652	0.547
Evia	0.286	0.327	0.312	0.279	0.319	0.663	**0.380**
Montiferru	0.376	0.392	0.416	0.368	0.485	0.582	**0.405**

S3.4 The contrast pair

Manavgat and Muğla, 306 km apart in one country and one fire year, have the most similar burned-cell environmental range of any pair, and Bejís and Montiferru the least similar (Fig. 8; Table S7). For Manavgat and Muğla as drawn, seven of nine feature-response directions point opposite ways, and all six features supported in both regions have opposite signs, elevation included. On the collar the elevation reversal remains, at 0.376 [0.300, 0.465] in Manavgat against 0.606 [0.525, 0.685] in Muğla. Transfer is below chance in both directions on both frames, 0.438 and 0.345 as drawn and 0.493 and 0.433 under the collar, where the interval of Muğla to Manavgat, [0.381, 0.491], excludes chance. The most similar pair is among the weakest in the matrix but not the weakest: on the collar the weakest direction is Manavgat to Bejís at 0.407 and the strongest Muğla to Evia at 0.728. Bejís and Montiferru hardly overlap on any measure and transfer above chance in both directions, though at the point estimate only. On these two pairs high niche overlap did not ensure transfer and low overlap did not prevent it; the correlation across all pairs is tested in Section S4.5.

Table S7. The most and least similar pairs, on both frames. Schoener’s *D* uses burned cells only and does not change with the collar. Transfer values are the two ordered directions of a pair, and ranks are out of the twenty directions on the collar. *D* and transfer as drawn from `round5/out_official/figure_contrast_pairs.json` (the source of Fig. 8), collar transfer and ranks from `round5/collar/aoi_frame_transfer.csv`, and the AoA shares from

round5/collar/aoa_directed_pair_summary.csv; paper/figures/fig8_contrast_pairs.py checks every cell.

	Manavgat and Muğla	Bejís and Montiferru
Schoener's *D*, mean 1-D	**0.799** (highest)	**0.479** (lowest)
per-feature *D*	0.74 to 0.87	0.23 to 0.77
transfer, frames as drawn	0.438, 0.345	0.594, 0.548
transfer, 10 km collar	0.493, 0.433	0.669, 0.624
rank of 20 on the collar, from the bottom	4th, 2nd	16th, 12th
target cells inside the AoA	0.876, 0.531	n/a

S3.5 The two Muğla fires

Muğla burned in 2021 and again eleven months later, on the same grid, so this pair holds the place fixed and changes the fire. It is reported only here. The 2022 population is the 2021 population without the 2021 scar (38,790 cells with 331 burned, against 41,730 with 2,911), the two share 38,789 cells, and the 2022 fire started about six weeks earlier in its season, so year and seasonal phase are confounded. As drawn, elevation reverses with bootstrap support, from 0.611 [0.532, 0.690] in 2021 to 0.296 [0.230, 0.355] in 2022, and the four absolute thermal channels cross 0.5 at the point estimate (current LST 0.325 to 0.515) while the two anomaly channels do not (LST anomaly 0.485 to 0.380; paper/step9g_raw/step9g_univariate_feature_auc_direction_reversal/mugla_2021__mugla_2022_event_relative/). The collar removes the elevation reversal. The 2022 scar is compact, so 93.2 % of the 2022 cells lie more than 10 km from it, and on the collar the 2022 value moves to 0.565, on the 2021 side of 0.5, with an interval that includes chance (code/mugla_two_event_collar.csv, paper/code/verify_mugla_collar.py). Only elevation and slope can be evaluated on the collar, because the modelling export of the 2022 fire is not in this repository and the rebuilt 2022 data differ from it on every seasonal channel.

S4 Transfer, adaptation and similarity

S4.1 The transfer matrix

Table S8 gives all twenty directions with and without label-free adaptation, as plotted in Figs. 4 and 5. At 2-cell blocking eleven directions are above chance with interval support and seven below: both directions between Manavgat and Bejís, between Manavgat and Muğla and between Bejís and North Evia, and Montiferru to Manavgat. At 10-cell blocking the same point estimates give 9 above, 6 below and 5 uncertain; Manavgat to Muğla loses its support, and no direction changes side of chance. The paired thermal contribution goes from twelve positive, seven negative and one uncertain verdict at 1 km to six, five and nine at 5 km, so coarser blocking removed support from eight verdicts and added none. Across five bootstrap seeds every 1 km verdict is stable, whereas at 5 km three change (the level verdict of Evia to Bejís and the paired verdicts of Manavgat to Muğla and Evia to Montiferru), which makes the counts of supported directions the least stable numbers of the paper (round5/out_official/transfer_ci_blocksize.csv, round6/seed_stability.json). The random-forest seed is fixed, so every interval is conditional on one fitted source model; Table S1 gives the spread over ten seeds.

Under region-wise z-scoring the twenty directions range from 0.302 to 0.630, and under CORAL from 0.406 to 0.624. With the better of the two methods in each direction, 15 directions end closer to chance and 5 further from it, four Montiferru directions that move up and Manavgat to Muğla, which moves down from 0.438 to 0.427; under CORAL alone, 16 end closer (round5/matrix20_official.csv).

Table S8. The transfer matrix, thermal model, natural-vegetation population. Target ROC-AUC with 2-cell spatial-block bootstrap 95 % intervals (1000 replicates). CORAL follows region-wise z-scoring ($\lambda = 10^{-5}$). Generated by paper/code/table_b9.py from round5/matrix20_official.csv.

Direction	Raw	Region-wise z-score	CORAL
Manavgat→Bejís	0.396 [0.373, 0.422]	0.450 [0.423, 0.477]	0.467 [0.441, 0.491]
Bejís→Manavgat	0.314 [0.296, 0.332]	0.302 [0.282, 0.323]	0.406 [0.388, 0.423]

Direction	Raw	Region-wise z-score	CORAL
Manavgat→Muğla	0.438 [0.418, 0.456]	0.427 [0.408, 0.444]	0.417 [0.398, 0.436]
Muğla→Manavgat	0.345 [0.331, 0.359]	0.485 [0.468, 0.502]	0.476 [0.460, 0.493]
Manavgat→Evia	0.654 [0.633, 0.676]	0.529 [0.504, 0.553]	0.504 [0.481, 0.528]
Evia→Manavgat	0.677 [0.658, 0.697]	0.404 [0.386, 0.421]	0.417 [0.399, 0.435]
Bejís→Muğla	0.618 [0.601, 0.635]	0.518 [0.501, 0.535]	0.507 [0.489, 0.524]
Muğla→Bejís	0.583 [0.561, 0.607]	0.535 [0.512, 0.557]	0.560 [0.538, 0.581]
Bejís→Evia	0.383 [0.363, 0.402]	0.532 [0.509, 0.551]	0.499 [0.479, 0.518]
Evia→Bejís	0.448 [0.426, 0.470]	0.549 [0.524, 0.575]	0.549 [0.525, 0.573]
Muğla→Evia	0.653 [0.636, 0.671]	0.561 [0.543, 0.580]	0.563 [0.545, 0.582]
Evia→Muğla	0.577 [0.560, 0.593]	0.501 [0.485, 0.518]	0.530 [0.515, 0.546]
Montiferru→Manavgat	0.404 [0.385, 0.422]	0.388 [0.367, 0.408]	0.436 [0.416, 0.456]
Manavgat→Montiferru	0.518 [0.474, 0.563]	0.527 [0.484, 0.568]	0.505 [0.461, 0.547]
Montiferru→Bejís	0.548 [0.521, 0.578]	0.574 [0.552, 0.596]	0.569 [0.548, 0.591]
Bejís→Montiferru	0.594 [0.560, 0.631]	0.550 [0.500, 0.601]	0.574 [0.530, 0.621]
Montiferru→Muğla	0.619 [0.604, 0.634]	0.576 [0.562, 0.589]	0.565 [0.550, 0.579]
Muğla→Montiferru	0.531 [0.495, 0.568]	0.587 [0.549, 0.624]	0.584 [0.547, 0.623]
Montiferru→Evia	0.586 [0.565, 0.606]	0.630 [0.611, 0.649]	0.624 [0.605, 0.641]
Evia→Montiferru	0.647 [0.608, 0.682]	0.568 [0.528, 0.609]	0.581 [0.539, 0.623]

S4.2 The transfer-gap decomposition

Table S9 splits the gap between the within-region skill of the target and raw transfer into the part that the better label-free method recovers and the rest. Among the directions that started below chance the recovered fraction is at most +0.28 (Bejís to Evia). Seven directions have negative recovery, five of them with intervals below zero. The within-region reference is not matched to a transfer evaluation (Section 3.3), and the raw column depends on the study area (Section 3.4), so the fractions hold only within this protocol.

Table S9. Transfer-gap decomposition, four regions, 12 directions. Within = within-region thermal AUC of the target. Best adapted = the better of z-score and CORAL, a choice that uses target labels. Recovered fraction = (adapted − raw)/(within − raw), signed and not clipped, with a paired bootstrap interval (1000 replicates, 2-cell blocks). Status uses the 2-cell intervals of Table S8. Montiferru is not part of this decomposition. Source `four_aoi_decomposition.csv` in `round5/tables/corrected/`.

Direction	Within	Raw	Best adapted (method)	Recovered fraction [CI]	Status
Bejís→Evia	0.912	0.383	0.532 (z-score)	+0.28 [+0.23, +0.32]	adapted above chance
Muğla→Manavgat	0.908	0.345	0.485 (z-score)	+0.25 [+0.22, +0.28]	adapted, chance not excluded
Evia→Bejís	0.918	0.448	0.549 (z-score)	+0.22 [+0.16, +0.27]	adapted above chance
Bejís→Manavgat	0.908	0.314	0.406 (CORAL)	+0.15 [+0.13, +0.18]	adapted still below chance
Manavgat→Bejís	0.918	0.396	0.467 (CORAL)	+0.14 [+0.09, +0.17]	adapted still below chance
Manavgat→Muğla	0.859	0.438	0.427 (z-score)	−0.03 [−0.06, +0.01]	**negative recovery**
Muğla→Bejís	0.918	0.583	0.560 (CORAL)	−0.07 [−0.16, +0.01]	**negative recovery**
Evia→Muğla	0.859	0.577	0.530 (CORAL)	−0.17 [−0.22, −0.11]	**negative recovery**
Muğla→Evia	0.912	0.653	0.563 (CORAL)	−0.35 [−0.43, −0.27]	**negative recovery**
Bejís→Muğla	0.859	0.618	0.518 (z-score)	−0.42 [−0.51, −0.34]	**negative recovery**

Direction	Within	Raw	Best adapted (method)	Recovered fraction [CI]	Status
Manavgat→Evia	0.912	0.654	0.529 (z-score)	−0.48 [−0.56, −0.42]	**negative recovery**
Evia→Manavgat	0.908	0.677	0.417 (CORAL)	−1.13 [−1.27, −1.00]	**negative recovery**

S4.3 Precision and recall

A susceptibility map ranks areas, and at target prevalences of 7.0 to 28.7 % precision and recall can tell a different story from ROC-AUC. Table S10 gives PR-AUC against the no-skill baseline. The seven directions below their own baseline are the seven below chance on ROC-AUC. At 1 km blocking all seven have intervals below the baseline, and at 5 km two do, both into Manavgat: Bejís to Manavgat [0.067, 0.132] and Muğla to Manavgat [0.071, 0.135], against 0.143.

Table S10. Thermal transfer as PR-AUC against the no-skill baseline. The baseline is the burned prevalence of the target, and lift is PR-AUC divided by it, so a lift of 1 is a random ranking. Ordered by lift. From the step9b exports (`round5/tables/corrected/*/step9b_metrics.json`); intervals at 2-cell and 10-cell blocking in `round7/r7a_pr_auc_10cell.csv`.

Direction	ROC-AUC	PR-AUC	No-skill	Lift
Evia → Manavgat	0.677	0.321	0.143	**2.24**
Manavgat → Evia	0.654	0.407	0.287	1.42
Bejís → Montiferru	0.594	0.289	0.212	1.36
Evia → Montiferru	0.647	0.283	0.212	1.34
Bejís → Muğla	0.618	0.093	0.070	1.33
Muğla → Evia	0.653	0.379	0.287	1.32
Montiferru → Bejís	0.548	0.093	0.072	1.28
Montiferru → Muğla	0.619	0.089	0.070	1.27
Evia → Muğla	0.577	0.086	0.070	1.23
Muğla → Bejís	0.583	0.088	0.072	1.22
Manavgat → Montiferru	0.518	0.243	0.212	1.15
Montiferru → Evia	0.586	0.317	0.287	1.11
Muğla → Montiferru	0.531	0.214	0.212	1.01
Manavgat → Muğla	0.438	0.060	0.070	**0.85**
Evia → Bejís	0.448	0.059	0.072	**0.82**
Montiferru → Manavgat	0.404	0.113	0.143	**0.79**
Bejís → Evia	0.383	0.226	0.287	**0.79**
Manavgat → Bejís	0.396	0.054	0.072	**0.75**
Muğla → Manavgat	0.345	0.101	0.143	**0.70**
Bejís → Manavgat	0.314	0.098	0.143	**0.68**
Mean	**0.527**	**0.181**	**0.157**	**1.13**

S4.4 Pooled training

A model trained on the pooled populations of the other four regions (Fig. 6) beats the mean of the four single-source models for one target of five, Evia, at 0.715 [0.668, 0.757] against a mean of 0.569, and falls short of this mean for the other four targets by 0.009 to 0.057. It beats the best single source only for Evia (0.654), and that source can only be chosen with target labels. For every target the pooled model stays 0.20 to 0.48 AUC below the within-region ceiling, and for Manavgat and Bejís it is below chance, with interval support only for Manavgat, at 0.426 [0.369, 0.486] (`code/loro_all.json`, `round7/r7d_pooled_vs_single.csv`).

S4.5 Similarity measures

Twenty similarity measures in five families were each rank-correlated with raw thermal transfer over the directions on which they are defined, with one pair-based bootstrap (unordered pairs resampled with

replacement, both directions kept, 2000 replicates, seed 42). The marginal family compares predictor distributions: area-of-applicability dissimilarity in scaled, importance-weighted predictor space (Meyer and Pebesma 2021, 2022; Ludwig et al. 2023), climatic distance and a domain classifier. The niche-overlap family uses Schoener's *D* (Schoener 1968) and Warren's *I* (Warren et al. 2008) over burned cells, the regime family counts burned patches, and the conditional family measures how often signed associations agree, over all predictors and over those with interval support; geographic distance is the fifth family. The measures were recorded on 8 August 2026 with their results under the original label (commit abd6a4b on the `history-archive` branch of the public repository) and were not re-selected after the label correction. This is a dated record and was not pre-registered.

As drawn, no interpretable measure has an interval that excludes zero (Table S11). The conditional family comes closest, at +0.52 and +0.49; under the original label these two rows excluded zero (+0.84 and +0.81), and after the correction they do not. Geographic distance does not order the matrix, and the domain classifier separates source from target at AUC ≥ 0.96 for every pair, so it gives no ordering. Over all 120 permutations of the five region labels the smallest two-sided *p*-value is 0.10 (vector Spearman, six directions) and the two supported-feature measures give 0.125 and 0.15, where the smallest attainable value is 1/120 (`round8/r8d_summary.json`). On the twelve directions common to all families, every interpretable row still includes zero (`round5/out_official/diagnostics_common_subset.json`).

Ten independent region pairs give little power, so a moderate ordering would usually be missed. Measures built on interval support are also unstable: five to six of the pipeline's interval bounds lie within 0.01 of 0.5, and one such flag, the NDVI of Manavgat, moved the supported-feature cosine from ρ = 0.49 to 0.70 when two equally valid bootstrap streams disagreed on it.

Table S11. Similarity measures against raw thermal transfer. Spearman ρ with pair-based bootstrap 95 % intervals; Exp. = expected sign. Rows with n = 12 cover the four regions other than Montiferru, for which these measures were not produced, and the supported conditional rows use the 18 directions with at least one supported feature. Source `round5/s6_diagnostics_20.csv`. ‡ The last row belongs to the set fixed in advance and stays in the table, but it is defined on six directions and its interval is degenerate (the upper bound equals the point estimate), so it is not interpreted.

Diagnostic	Family	Exp.	n dir	Spearman ρ [95% CI]	CI excludes 0
Agreement fraction, supported features	P(y|x) conditional	+	18	+0.52 [−0.27, +0.87]	no
Cosine, supported features	P(y|x) conditional	+	18	+0.49 [−0.24, +0.87]	no
Cosine, all 9 features	P(y|x) conditional	+	20	+0.44 [−0.29, +0.80]	no
Vector Spearman, all 9	P(y|x) conditional	+	20	+0.43 [−0.21, +0.80]	no
Agreement count, all 9	P(y|x) conditional	+	20	+0.47 [−0.29, +0.80]	no
Schoener's D, 1-D mean	P(x|y=1) niche	+	20	+0.10 [−0.53, +0.63]	no
Warren's I, 1-D mean	P(x|y=1) niche	+	20	+0.19 [−0.43, +0.77]	no
Schoener's D, PCA-2D	P(x|y=1) niche	+	20	+0.03 [−0.59, +0.63]	no
Warren's I, PCA-2D	P(x|y=1) niche	+	20	−0.03 [−0.66, +0.56]	no
Mahalanobis, burned centroids	P(x|y=1) niche	−	20	−0.22 [−0.82, +0.42]	no
Domain-classifier AUC	P(ix) marginal	−	20	−0.33 [−0.81, +0.36]	no
Predictor-space mean dissimilarity	P(ix) marginal	−	12	−0.13 [−0.67, +0.37]	no
Predictor-space p95 dissimilarity	P(ix) marginal	−	12	−0.15 [−0.68, +0.24]	no
Fraction inside weighted AoA	P(ix) marginal	+	12	+0.21 [−0.48, +0.62]	no
Fraction inside unweighted support	P(ix) marginal	+	12	+0.08 [−0.92, +0.76]	no
Climatic distance	P(ix) marginal	−	12	+0.03 [−0.84, +0.84]	no
Geographic distance	geographic	−	12	−0.17 [−0.87, +0.84]	no
Regime distance, log effective-N	P(y) structure	−	20	+0.11 [−0.67, +0.80]	no

Diagnostic	Family	Exp.	n dir	Spearman ρ [95% CI]	CI excludes 0
Regime distance, largest share	P(y) structure	−	20	+0.11 [−0.66, +0.77]	no
Vector Spearman, supported (≥3 feats) ‡	P(y|x) conditional	+	6	+0.96 [+0.85, +0.96]	degenerate interval; not interpreted

S5 Data and methods

S5.1 Label, grid, gate and the Manavgat correction

Labels come from MODIS MCD64A1 Collection 6.1 (`MODIS/061/MCD64A1`) through Google Earth Engine (Gorelick et al. 2017). Each native ~500 m observation is copied onto the 30 m EPSG:4326 grid of the pipeline, and the analysis grid is rebuilt from it in blocks of `round(500 / 30) = 17 × 17` pixels, identified by `row_500m` and `col_500m`. The cell is square in degrees, about 510 m from north to south and 390 to 407 m from east to west (0.199 to 0.208 km², against 0.215 km² for the native cell), so it approximates the MODIS cell without reproducing it. Block sizes refer to the north-south edge, and the collar step of 0.45 km per cell lies between the two edges, so collar radii are nominal to within about 15 %. A cell counts as burned when one burned sub-pixel falls inside the label window, which can widen a scar by one cell at its edge, and its burn date is the modal sub-pixel day of year. The label never decides whether a cell is modelled.

Before any modelling, each region passes a gate. It is admitted as a wildfire case when at least 30 cells burned and at least half of the burned cells are dominated by natural vegetation (ESA WorldCover classes on the same cells), and rejected as a cropland control when the cropland share is at least half. The gate separates burning of natural fuel from stubble burning after harvest, which MCD64A1 does not distinguish; Table 1 gives its results.

Table S12 gives the earlier burning of each region. Burning inside the predictor window was removed by the pipeline except in Bejís, and burning in the five previous years is not screened anywhere; removing either changes little (Figs. S1 and S2). The only historical exclusion applied by design removes the 2021 Muğla scar from the 2022 population of Section S3.5.

Table S12. Earlier burning by region, natural-vegetation population. Pre-label burning lies inside the predictor window and historical burning in the five previous years. Recomputed from the MCD64A1 export of each region by `paper/code/ems_labels_1_prelabel.py` and `ems_labels_2_history.py` (`labels/r1_prelabel.json`, `labels/r2_history.json`).

Region	Pre-label cells removed	Pre-label burned cells retained	Historical-burn cells (five years)	… of which burned again in the event
Manavgat 2021	none needed (no pre-label burning)	0	0	0
Bejís 2022	exclusion not applied	48 (49 in all valid cells; 10 burned again)	163	0
Muğla 2021	49	0	194	25
North Evia 2021	16	0	366	88
Montiferru 2021	61	0	58	32

The label first used for Manavgat 2021 was exported on 8 July 2026, before the month-alignment fix of the MCD64A1 query in the pipeline (commit 183be42, 11 July), and was not renewed, so it missed the first four days of the fire, 28 to 31 July; the code itself was correct. The corrected label only adds burned cells (784 to 2,935 in the primary population) and changes no predictor or validity flag. With it the Manavgat rows of Table 2 become stronger: absolute AUCs rise by 0.04 to 0.11, and the 1 km interval narrows from [+0.055, +0.079] to [+0.060, +0.073], because 814 blocks of 2 cells now hold burned cells instead of 235. The outputs were re-frozen with the pipeline at commit 6381f4c, Python 3.12.10 and scikit-learn 1.9.0, with the two

changes given in Section S5.7. A control run re-froze the original label with the same code and reproduced the published outputs to within 10^{-5}, apart from the exceptions listed in the manifest, so differences between the runs come from the label alone. The correction, the re-freeze and the control run were carried out by the corresponding author with code written with Claude (see the statement on the use of generative AI in the main article) and were not repeated by the author of the pipeline; the manifest, hashes and scripts are in `paper/data/manavgat_2021/refreeze/`.

S5.2 Study areas and windows

Table S13 gives the bounding box and the windows of each study area. The 2022 Muğla windows are set relative to the 2022 fire with the 2021 lengths: a 58-day predictor window that ends the day before ignition and a 49-day label window that opens on it. The Evia rectangle was extended after the first one showed a very high burned fraction, and Section S1.2 gives transfer on the first rectangle.

Table S13. Study regions, study areas and time windows. Bounding boxes in EPSG:4326, as recorded. Day counts include both ends. The baseline years are the four years before each predictor window, on the same calendar dates.

Region	Bounding box (lon min, lat min, lon max, lat max)	Predictor window	Label window	Baseline years
Manavgat 2021 (Türkiye)	31.05, 36.72, 31.85, 37.35	2021-06-01 to 2021-07-27 (57 d)	2021-07-28 to 2021-08-31 (35 d)	2017, 2018, 2019, 2020
Bejís 2022 (Spain)	-1.05, 39.68, -0.35, 40.15	2022-06-15 to 2022-08-14 (61 d)	2022-08-15 to 2022-09-30 (47 d)	2018, 2019, 2020, 2021
Muğla 2021 (Türkiye)	27.1, 36.6, 28.9, 37.45	2021-06-01 to 2021-07-28 (58 d)	2021-07-29 to 2021-09-15 (49 d)	2017, 2018, 2019, 2020
North Evia 2021 (Greece)	23.05, 38.55, 23.85, 39.15	2021-06-05 to 2021-08-02 (59 d)	2021-08-03 to 2021-09-30 (59 d)	2017, 2018, 2019, 2020
Montiferru 2021 (Italy)	8.45, 40.05, 8.75, 40.27	2021-05-25 to 2021-07-23 (60 d)	2021-07-24 to 2021-08-31 (39 d)	2017, 2018, 2019, 2020

S5.3 Predictors

Ten predictors are summarised per cell over the predictor window (Fig. 2; Table S14). NDVI, current LST, the LST anomaly and the two TVDI channels come from Landsat 8 Collection 2 Level-2, screened per pixel with `QA_PIXEL` (the water bit is kept); elevation and slope come from the Copernicus DEM GLO-30, whose EGM2008 geoid heights are used without datum conversion; land cover is the dominant ESA WorldCover v200 class (Zanaga et al. 2022). The LST anomaly is a z-score of the current LST median against the four baseline years, set to no-data where the baseline standard deviation is below 1.0 °C. TVDI (Sandholt et al. 2002) uses wet and dry edges at the 2nd and 98th LST percentiles within twenty NDVI bins, fitted for each study area and window, so a TVDI of 0.5 means a different moisture state in each region; the TVDI difference is its raw anomaly against the same four years. Downscaled LST comes from a MODIS-to-Landsat model that takes coordinates as inputs, the only route by which coordinates enter a feature set (Fig. S1 gives the gain without it), and fused LST keeps the observed Landsat LST wherever it is valid and fills 0.11 % to 9.70 % of cells, by region, from the downscaled surface.

The six thermal channels carry about two signals. Current, fused and downscaled LST correlate at 0.97 to 1.00 in every region, current LST and TVDI at 0.88 to 0.98, and the two anomaly channels at 0.71 to 0.94, and two principal components explain 92 % to 98 % of their variance (`labels/r4_dimensionality.json`). The anomaly channels rest on one to seven clear Landsat observations per pixel and year (Table S15), which adds noise to the four-year baseline.

Table S14. Input data. All retrieved through Google Earth Engine.

Input	Product and version	Native resolution	Use	Reference
Burned area	MODIS MCD64A1 Collection 6.1 (`MODIS/061/MCD64A1`)	500 m, monthly	label	(Giglio et al. 2018, 2021)

Input	Product and version	Native resolution	Use	Reference
Surface reflectance and temperature	Landsat 8 Collection 2 Level-2 (`LANDSAT/LC08/C02/T1_L2`), `QA_PIXEL` screening, water bit preserved	30 m	NDVI, current LST, anomaly, TVDI	(U.S. Geological Survey Earth Resources Observation and Science (EROS) Center 2020)
Coarse LST	MODIS MOD11A1 v061 (`MODIS/061/MOD11A1`)	1 km, daily	downscaled and fused LST (gap-fill 0.11 % to 9.70 %)	(Wan et al. 2021)
Terrain	Copernicus DEM GLO-30 (EGM2008 heights)	30 m	elevation, slope	(European Space Agency 2022)
Land cover	ESA WorldCover v200 (2021)	10 m	dominant class, gate, population	(Zanaga et al. 2022)

Table S15. Landsat 8 observations per window. Scenes that intersect the study area, and clear observations per pixel after the pipeline's QA mask (`src/step3_landsat_lst.py`), median and 10th to 90th percentile at 300 m. The pipeline also filters by calendar month, so its counts can be slightly lower; Landsat 9 is not used. Source `round8/r8k_landsat_observations.csv`.

Region	Predictor window scenes	Clear observations, predictor window	Clear observations, baseline years (medians)
Manavgat 2021	14	6 [3 to 10]	7, 7, 6, 6
Bejís 2022	16	3 [2 to 6]	4, 4, 3, 2
Muğla 2021	18	6 [3 to 10]	6, 6, 6, 6
North Evia 2021	8	4 [3 to 4]	4, 3, 2, 1
Montiferru 2021	4	3 [2 to 4]	4, 3, 2, 3

S5.4 Resampling units

Uncertainty comes from a spatial-block bootstrap. With $\beta_1, \ldots, \beta_M$ the blocks of Eq. (1) of the main text that contain cells of F, replicate b draws M indices u_{bm} uniformly with replacement:

$$F^{*b} = \biguplus_{m=1}^{M} \beta_{u_{bm}}, \qquad \mathrm{CI}_{95} = [Q_{0.025}\{\theta(F^{*b})\}_b,\ Q_{0.975}\{\theta(F^{*b})\}_b]. \qquad (S1)$$

Here θ is the statistic and Q gives the 2.5 and 97.5 percentiles over 1000 replicates, seed 42, unless another number is given. A replicate with one class only is dropped, and differences are formed within each replicate, so they are paired. The reliability of an interval depends on the number of blocks with at least one burned cell, which falls from 192 to 843 at 2 cells to 6 to 33 at 20 cells. The 10-cell blocking, with 16 to 70 such blocks per region, is the coarsest that the design supports, and the 20-cell row of Table 2 is indicative.

Each region appears in eight of the twenty directions, so the *pair-cluster bootstrap* resamples the ten unordered region pairs, each carrying its set π_p of ordered directions:

$$\bar{\delta}^{*b} = \frac{\sum_{m=1}^{10} \sum_{(s,t)\in\pi_{u_{bm}}} \delta_{st}}{\sum_{m=1}^{10} |\,\pi_{u_{bm}}|}. \qquad (S2)$$

Clustering by target region replaces π_p by the four directions that share a target. A quantity with one value per held-out scar or target region gets a Student *t* interval over its n units:

$$\bar{x} \pm t_{0.975,\,n-1}\, s_x/\sqrt{n}. \qquad (S3)$$

The effective sample is therefore ten pairs or five regions for direction-level intervals, and at most seven scars in three regions for scar-level intervals. An assertion in the code excludes the same columns from every model fit: coordinates (`lon`, `lat`, `row`, `col` and their normalised forms), every burn-date and label-provenance column, and the agreement fraction.

S5.5 Gap decomposition and reversal criterion

The recovered fraction of a direction is (adapted − raw) / (within − raw), signed and not clipped, with its interval from the same paired bootstrap. It describes what the two tested methods achieved, the better one chosen with target labels, and much of the remainder already appears inside one region (Section 3.3), so it is not a concept-shift residual. The signed univariate association is the raw ROC-AUC of a predictor against `burned`, not folded to max(AUC, 1 − AUC). A reversal is bootstrap-supported only when the two point estimates lie on opposite sides of 0.5 and the 10-cell interval of each region excludes 0.5 (Section S3.1); `conditional_similarity_transfer.json` stores the rule as metadata. Replicates with $|G| < 10^{-6}$ are dropped.

S5.6 Label-free adaptation

Region-wise z-scoring standardises the numeric features of each region with the statistics of that region, source statistics from source data and target statistics from target data, never pooled. For feature j in region R,

$$z_{ij} = \frac{x_{ij} - \mu_j^R}{\sigma_j^R}, \qquad \mu_j^R = \frac{1}{n_j^R}\sum_{i \in O_j^R} x_{ij}\,, \qquad \sigma_j^R = \left(\frac{1}{n_j^R}\sum_{i \in O_j^R} (x_{ij} - \mu_j^R)^2\right)^{1/2}, \qquad (S4)$$

where O_j^R contains the n_j^R cells with an observed value (ddof 0). A missing value is first set to μ_j^R, so it becomes zero, and $\sigma_j^R < 10^{-12}$ is replaced by 1; land cover is not transformed. The classifier is refitted on the z-scored source and applied to the z-scored target, which removes differences in the mean and variance of each feature, so the adapted models use target feature statistics and never target labels.

CORAL then aligns the source covariance to the target covariance with the standard whitening and recolouring map (Sun et al. 2016). With Z_R the $n_R \times d$ matrix of z-scored numeric features, one row per cell,

$$Z_s^{\mathrm{al}} = Z_s\,(C_s + \lambda I)^{-1/2}\,(C_t + \lambda I)^{1/2}, \qquad C_R = \frac{1}{n_R}\sum_{i=1}^{n_R}(z_i - \bar{z}_R)(z_i - \bar{z}_R)^{\top}, \qquad (S5)$$

with $\lambda = 10^{-5}$ throughout (ddof 0). Both means are zero after Eq. (S4), so the mean terms of the general map are zero, and matrix powers use a symmetric eigendecomposition with eigenvalues set to at least 10^{-12}. The transform is applied to the source only; the target stays at Z_t, and the classifier is refitted on Z_s^{al}. A run-time check confirms that neither method uses a target label. Section S1.2 gives the effect of other values of λ.

S5.7 Reproduction, code and data

The transfer analysis runs in an environment separate from the pipeline, so the repository refits every within-region model and all twenty CORAL transfers and compares them with the frozen pipeline output, for Manavgat the output re-frozen at pipeline commit 6381f4c (Section S5.1). That re-freeze needed two changes: a one-line patch that lets the window-closure module accept a population column added by the corrected label (its diff is released with the re-freeze), and a runtime setting that points the Earth Engine calls at the project `thermaltwin`. Within-region results agree exactly and the transfer directions to within 1.3×10^{-8}, below the tolerance of 10^{-6}. The frame-transfer script of Section 3.4 fits its forests in parallel, so its values vary between runs by up to 2×10^{-6}, within the tolerance of 10^{-5} applied to it. Point estimates depend on the scikit-learn version: under 1.5.2 instead of 1.9.0, single directions move by up to 0.023 for the thermal model and 0.047 for the baseline, means by at most 0.003, and one direction changes interval support on each frame (`round7/sklearn152/compare_vs_1_9_0.json`).

The analysis code is public at https://github.com/dryuemco/thermal-twin under the MIT licence, and the version used here is archived at Zenodo (https://doi.org/10.5281/zenodo.23070876); the README of the repository describes each folder. The satellite processing pipeline is a separate public repository, https://github.com/emrehann17/satellite-thermal-digital-twin (MIT licence). The environment is Python 3.12.10 with NumPy 2.4.4, pandas 3.0.2 and scikit-learn 1.9.0, fixed in `ENVIRONMENT.md`. The five modelling datasets are in `paper/data/<region>/step8a_500m_modeling_dataset.parquet`, each checked on load against the SHA-256 in `paper/code/_canonical.py`; for Manavgat this is the corrected

re-freeze (`5a5e876c…`), while the pipeline's own record still names the original-label table (`054a1961…`), which the pipeline can regenerate. All satellite inputs are public (Table S14), and the data derived from them remain under the terms of their products.

References

European Space Agency (2022) Copernicus DEM GLO-30 [dataset]. European Space Agency. https://doi.org/10.5270/ESA-c5d3d65

Giglio L, Boschetti L, Roy DP, et al (2018) The Collection 6 MODIS burned area mapping algorithm and product. Remote Sensing of Environment 217:72–85. https://doi.org/10.1016/j.rse.2018.08.005

Giglio L, Justice C, Boschetti L, Roy D (2021) MODIS/Terra+Aqua burned area monthly L3 global 500m SIN grid V061 [dataset]. NASA Land Processes Distributed Active Archive Center. https://doi.org/10.5067/MODIS/MCD64A1.061

Gorelick N, Hancher M, Dixon M, et al (2017) Google Earth Engine: Planetary-scale geospatial analysis for everyone. Remote Sensing of Environment 202:18–27. https://doi.org/10.1016/j.rse.2017.06.031

Ludwig M, Moreno-Martinez A, Hölzel N, et al (2023) Assessing and improving the transferability of current global spatial prediction models. Global Ecology and Biogeography 32:356–368. https://doi.org/10.1111/geb.13635

Meyer H, Pebesma E (2021) Predicting into unknown space? Estimating the area of applicability of spatial prediction models. Methods in Ecology and Evolution 12:1620–1633. https://doi.org/10.1111/2041-210X.13650

Meyer H, Pebesma E (2022) Machine learning-based global maps of ecological variables and the challenge of assessing them. Nature Communications 13:2208. https://doi.org/10.1038/s41467-022-29838-9

Sandholt I, Rasmussen K, Andersen J (2002) A simple interpretation of the surface temperature/vegetation index space for assessment of surface moisture status. Remote Sensing of Environment 79:213–224. https://doi.org/10.1016/S0034-4257(01)00274-7

Schoener TW (1968) The Anolis lizards of Bimini: Resource partitioning in a complex fauna. Ecology 49:704–726. https://doi.org/10.2307/1935534

Sun B, Feng J, Saenko K (2016) Return of frustratingly easy domain adaptation. In: Proceedings of the thirtieth AAAI conference on artificial intelligence (AAAI-16). AAAI Press, Phoenix, Arizona, USA, pp 2058–2065

U.S. Geological Survey Earth Resources Observation and Science (EROS) Center (2020) Landsat 8-9 operational land imager / thermal infrared sensor level-2, collection 2 [dataset]. US Geological Survey. https://doi.org/10.5066/P9OGBGM6

Wan Z, Hook S, Hulley G (2021) MODIS/Terra land surface temperature/emissivity daily L3 global 1km SIN grid V061 [dataset]. NASA Land Processes Distributed Active Archive Center. https://doi.org/10.5067/MODIS/MOD11A1.061

Warren DL, Glor RE, Turelli M (2008) Environmental niche equivalency versus conservatism: Quantitative approaches to niche evolution. Evolution 62:2868–2883. https://doi.org/10.1111/j.1558-5646.2008.00482.x

Zanaga D, Van De Kerchove R, Daems D, et al (2022) ESA WorldCover 10 m 2021 v200. Zenodo. https://doi.org/10.5281/zenodo.7254221